

will build on the ideas presented here: deterministic, mathematical formulas that can resemble randomness as evaluated by stringent mathematical tests.

Summary

Mathematics and algorithms will always have a close relationship. The more deeply you dive into one field, the more ready you will be to take on advanced ideas in the other. Math may seem arcane and impractical, but it is a long game: theoretical advances in math sometimes lead to practical technologies only many centuries later. In this chapter we discussed continued fractions and an algorithm for generating continued fraction representations of any number. We also discussed square roots, and examined an algorithm that handheld calculators use to calculate them. Finally, we discussed randomness, including two algorithms for generating pseudorandom numbers, and mathematical principles that we can use to evaluate lists that claim to be random.

In the next chapter, we will discuss optimization, including a powerful method you can use to travel the world or forge a sword.